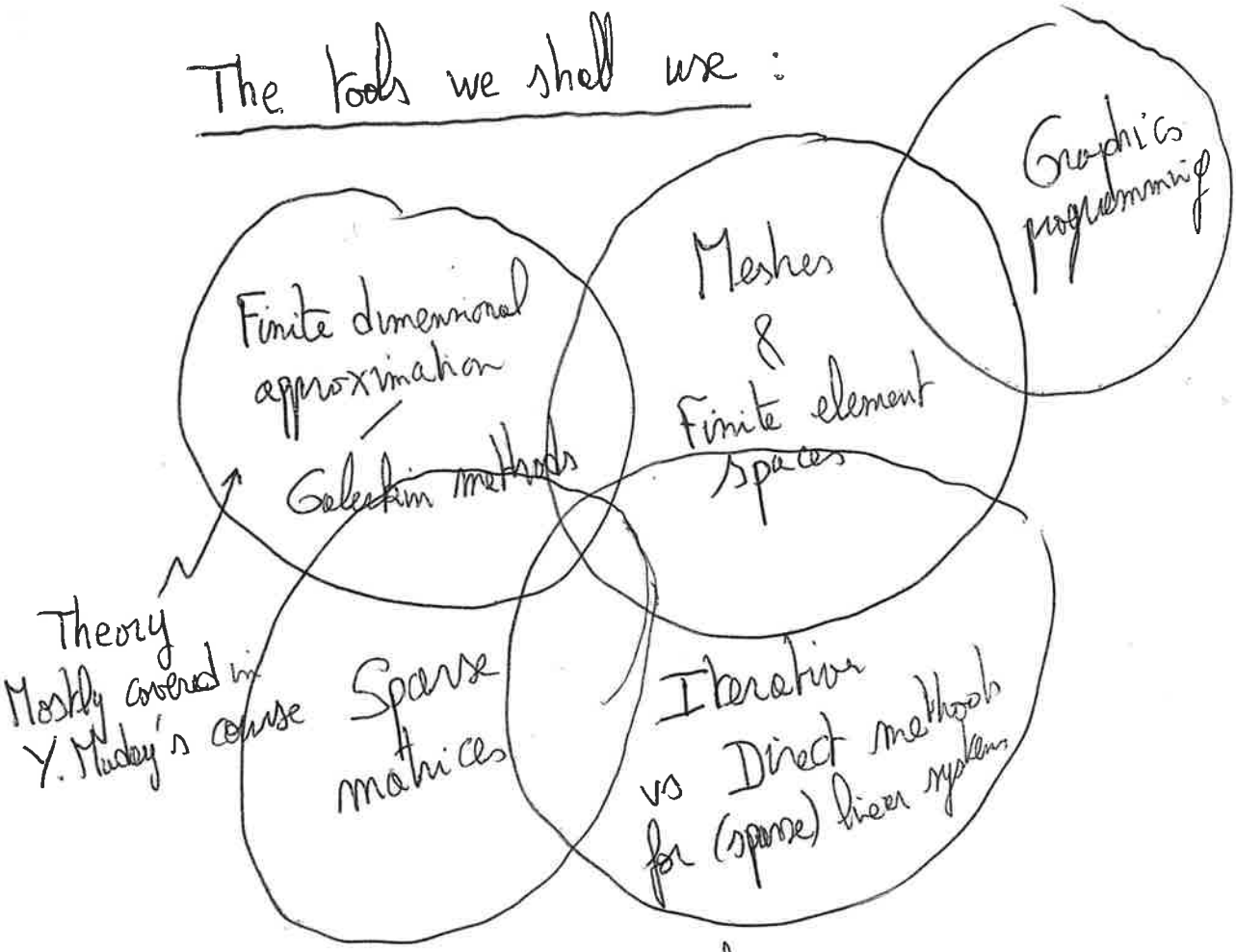
Goal: Implement a finite element strategy from scratch to solve a few representative problems including:

- a Poisson solver
- a Navier-Stokes solver.

The tools we shall use:



An "unusual" framework:

- 1D covered in M1 courses and too specific to represent real world applications
- 3D ~~quickly turns into~~ too big of a step for a ~~M1~~ course

2D Surface meshes in 3D

- 3D coordinates
- complexity of 2D
- interesting situations without the pain of dealing with bd.

<u>How "we" work</u> : • Concepts / Instruction explained "on the wall" at begining of lecture

• Actual work done on the computer with a <u>guided process</u>

↳ e.g. fill "holes" in proposed implementation files.

• Collaborative (each student pushes his own work to a branch of a github repo)

• Proposed solutions are uploaded regularly

<u>Example 1</u> (Modified) Poisson solver on S^2.

① Theory (brief)

$$-\Delta_{S^2} u + u = f \quad \text{on } \Omega = S^2 \subseteq \mathbb{R}^3$$

Weak formulation

$$\int_\Omega \nabla u \cdot \nabla v + u v = \int_\Omega f v \quad \forall v \in H^1(\Omega)$$

i.e $\quad a(u, v) = \ell(v) \quad \forall v \in V$

where $a(u,v) := \int \nabla u \nabla v + uv$

$$\ell(v) = \int f v$$

<u>Note</u> . $a(u,v) \leq M \|u\| . \|v\| \quad \forall u,v \in V$ (continuity)

$$a(u,u) \geq \alpha \|u\|^2 \quad \forall u \in V \text{ (coercivity)}$$

(Here $\alpha = M = 1$ since $a(u,v) = \langle \, , \, \rangle_V$).

. If $f \in L^2(\Omega)$ [or even $H^{-1}(\Omega) = H^1(\Omega)'$], $|\ell(v)| \leq \|\ell\| . \|v\|$

Lax-Milgram (here even Riesz)

$$\exists ! \, u \in V \text{ s.t. } a(u,v) = \ell(v) \quad \forall v \in V$$

$$\text{and } \|u\| \leq \frac{\|\ell\|}{\alpha}.$$

② Finite dimensional approximation (needed for actual computations!)

Let $V_h \subset V$ a finite dimensional subspace

. "h" is an historical notation, h refers to a "step size" or "typical element size" of a discretization.

Lax-Milgram applies as well in V_h

$$\rightsquigarrow \exists ! \, u_h \in V_h \text{ s.t. } a(u_h, v_h) = \ell(v_h), \quad \forall v_h \in V_h$$

Suppose we have a basis $\varphi_1, \dots, \varphi_N$ of V_h. $\sim \dim(V_h)$

Write $u_h = \sum_{i=1}^{N} \alpha_i \varphi_i$

Then using $v_h = \varphi_j \quad (1 \le j \le N)$

we obtain the (square) linear system

$$A \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_N \end{pmatrix} = \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_N \end{pmatrix} := \begin{pmatrix} \ell(\varphi_1) \\ \vdots \\ \ell(\varphi_N) \end{pmatrix}$$

where $A = \left(a(\varphi_i, \varphi_j) \right)_{1 \le i,j \le N}$.

Q How can we insure that u_h is "close" to u ?

<u>Lemma</u> (Céa) Under the Lax-Milgram assumptions

$$\|u - u_h\| \le \frac{M}{\alpha} \underbrace{\inf_{v_h \in V_h} \|u - v_h\|}_{=: d(u, V_h)}$$

i.e error $\lesssim$ best approximation error in V_h

↳ shall be small if "h is small"

(quantitative versions can be explicitly stated - Bramble-Hilbert lemma)

<u>Def</u> A sequence $(V_n)_{n\in\mathbb{N}}$ of (finite dim) subspaces is called a conformal approximation sequence if

$$\forall u \in V, \ \exists \text{ sequence } v_n \in V_n \text{ s.t. } \|u - v_n\| \to 0$$

as $n \to +\infty$.

<u>Rmk</u>
- In typical applications the V_n need not be an ↗ sequence (not nec. nested)
- Conformal refers to the fact that $V_n \subseteq V$. In real world applications it often arises that $V_n \not\subseteq V$, in particular $\|u - u_h\|$ has no meaning.

$$I_h : V \longrightarrow V_h \quad \text{interpolation op.}$$

$$L_h : V_h \longrightarrow V \quad \text{lifting op.}$$

s.t. (often) $I_h \circ L_h = Id_{V_h}$

$$+ \ \|Id_V - L_h \circ I_h\| \to 0 \quad \text{as } h \to 0.$$

Should also distinguish
$$a : V \times V \to \mathbb{R}$$
$$\ell : V \to \mathbb{R}$$
$$a_h : V_h \times V_h \to \mathbb{R}$$
$$\ell_h : V_h \to \mathbb{R}$$

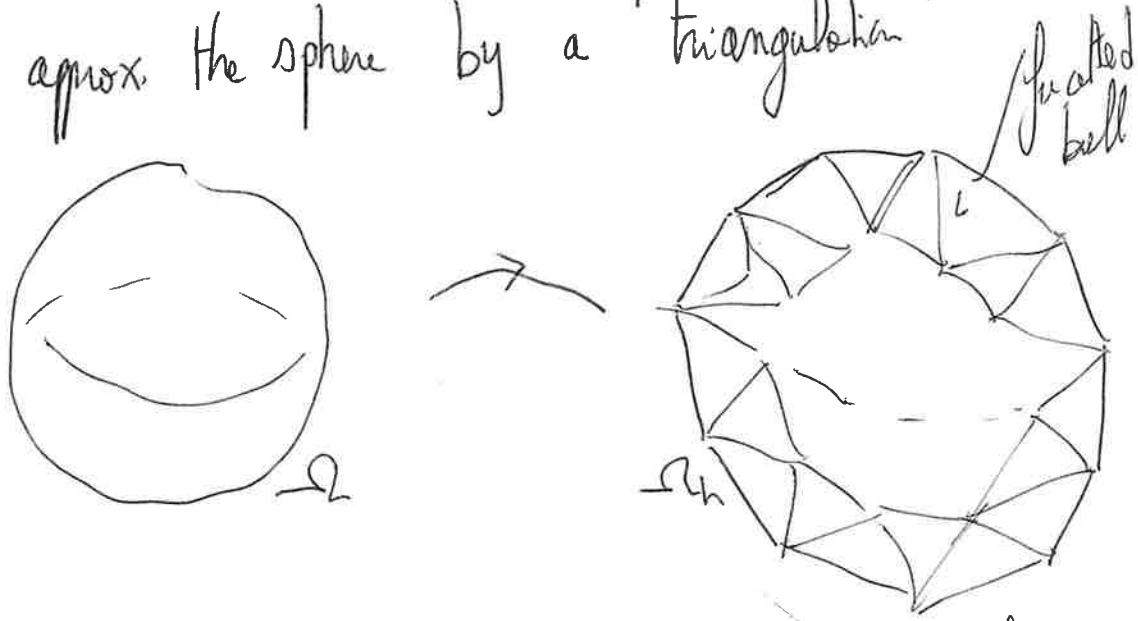
and measure their proximity.

<u>Ex</u> Extend Céa's lemma to a non conformal framework.

3 Effective finite dimensional approximation by <u>finite elements</u>. The simplest case of Lagrange $\mathbb{P}_1$ elements.

TL/DR: piece-wise affine functions, globally continuous, on a mesh.

- approx. the sphere by a "triangulation"



- $V_h \equiv \{ f : \Omega_h \to \mathbb{R}$ affine on each triangle, and globally continuous $\}$

<u>completely determined</u> by its value on each <u>vertex</u> of the mesh.

In general: motion of <u>degrees of freedom</u>, here equal to the vertices of the mesh.

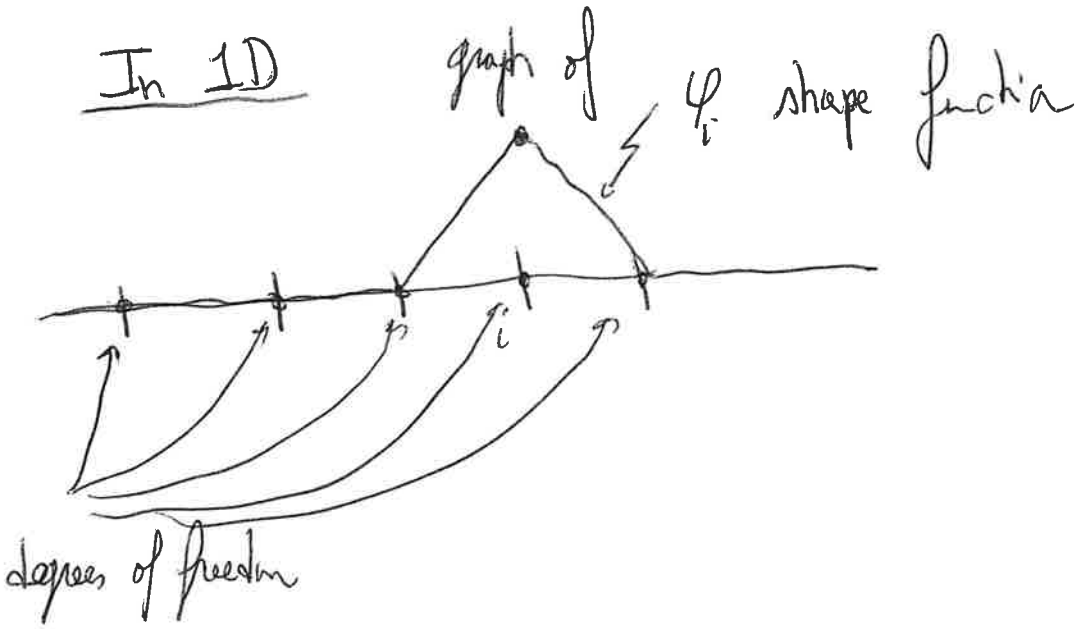
Associate to the degrees of freedom are the

so called <u>shape functions</u> (# shape = # degrees = $\dim(V_h)$)

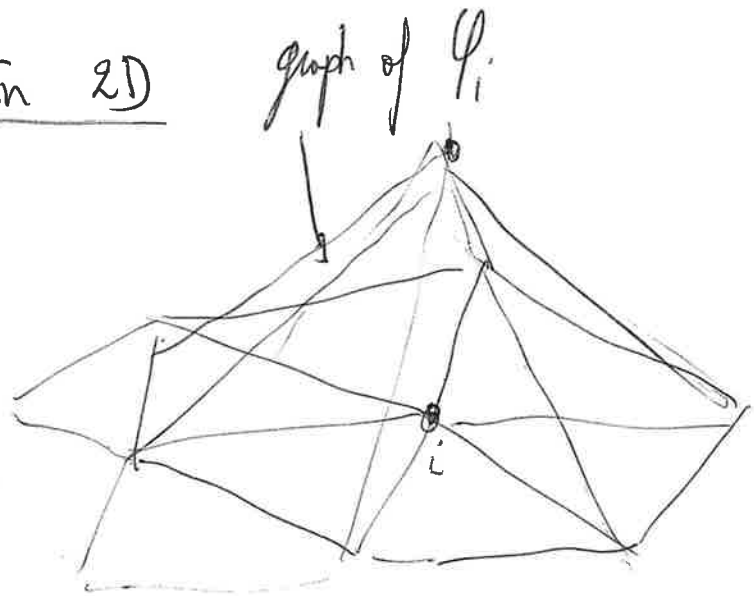
$\hookrightarrow$ fix one degree of freedom $= 1$

others $= 0$

<u>In 1D</u>



<u>In 2D</u>



<u>Important rmk</u> $\mathrm{supp}(\varphi_i) \cap \mathrm{supp}(\varphi_j) \neq \emptyset$

iff i and j are neighbours in the mesh.

Recall from lecture 1 :

We approximate the problem

$$-\Delta_{S^2} u + u = f$$

by a linear system

$$(S + M)\, U = MF$$

where $U = \begin{pmatrix} U_1 \\ \vdots \\ U_N \end{pmatrix}$, $F = \begin{pmatrix} F_1 \\ \vdots \\ F_N \end{pmatrix}$

called "Stiffness" matrix $\longrightarrow$ $S = \left(\int_{\Omega_h} \nabla \varphi_i \nabla \varphi_j \right)_{1 \le i,j \le N}$

"Mass" matrix $\longrightarrow$ $M = \left(\int_{\Omega_h} \varphi_i \varphi_j \right)_{1 \le i,j \le N}$

Ω_h triangular manifold mesh approximating S^2 with vertex set $\mathcal{V}_h$ and triangle set $\mathcal{T}_h$ $\left(\Omega_h = \bigcup_{T \in \mathcal{T}_h} T\right)$

$\# \mathcal{V}_h = N$ and $(\varphi_i)_{1 \le i \le N}$ the associate $\mathbb{P}_1$- Lagrange basis functions.

$$u_h = \sum_{i=1}^{N} U_i \varphi_i \qquad F_i = f\left(\frac{v_i}{\|v_i\|}\right)$$

corresponding vertex in V_h

We momentarily skip the actual $\left\{\begin{array}{l}\text{computations}\\ \text{construction}\end{array}\right.$ of M and S, this is part of <u>your</u> exercise for lecture 3.

It has the canonical form

$$Ax = b$$

with $A = S + M$, $b = MF$

$x = U$

(1) By construction A is symmetric, and it is also positive definite (it wouldn't without M because constant vectors would be in the kernel).

$$\sigma(A) \subseteq \cancel{]0,+\infty[}\ [\lambda_{min}, \lambda_{max}] \subseteq]0,+\infty[$$

Condition n^{b} $K(A) := \|A\| \cdot \|A^{-1}\| = \dfrac{\lambda_{max}}{\lambda_{min}} \geq 1.$

② By construction A is also sparse:

$A_{ij} \neq 0$ only if the corresponding v_i, v_j are neighbours in the mesh.

If each vertex has at most d_{max} neighbours

$$nnz(A) \leq (d_{max}+1)N$$

<u>Typical cases to keep in mind</u>

a) A triangular manifold mesh without boundary has by Euler formula

$$\underbrace{\#(F)}_{\text{faces} \equiv \text{triangles}} - \underbrace{\#(E)}_{\text{edges}} + \underbrace{\#(V)}_{\text{vertices}} = 2(1-g)$$

$= 2$ for a sphere

- Each edge is shared between 2 faces
- Each face has 3 edges

$$\Rightarrow 3\#(F) = 2\#(E)$$

Hence $\underbrace{\#(F)}_{n_{tri}} = 2\#(V) - 4$

$$= 2\underbrace{N}_{n_{vert}} - 4$$

b)

Computation on estimates for λ_{min} and λ_{max} on Ω_h

<u>Assumptions</u> length (edges) $\leq C h$ $(C>0)$

radius (inscribed circle) $\geq c h$ $(c>0)$

$\Rightarrow \nabla \varphi_i \simeq \frac{1}{h}$ on supporting triangles

- - -

<u>Rmk</u> It is typical for applications to allow a certain (sometimes solution depending) gradation in element sizes. For their analysis it is useful to consider scalar products on $\mathbb{R}^N$ that are adapted (i.e. weighted by domain of influence area)

<u>Conclusion</u> $\kappa(S+M) \simeq O(h^{-2})$

(and $N = O(h^{-d})$ $d \equiv$ dimension of the mesh

so $\kappa = O(N^{2/d})$

$= O(N)$ for surface mesh.

Gradient method

$A \in \mathbb{R}^{N\times N}$ SPD

$$Ax_* = b \iff x_* = \operatorname{argmin}\left\{ x \xmapsto{\Phi} \tfrac{1}{2}\langle Ax, x\rangle - \langle b, x\rangle \right\}$$

Let $x \in \mathbb{R}^N$, set

$$e = e(x) = x_* - x \quad \text{(error)}$$

$$r = r(x) = b - Ax \quad \text{(residual)}$$

Note: $Ae = r$.

Compute:

$$d\Phi(x)(h) = \lim_{\varepsilon\to 0} \frac{\Phi(x+\varepsilon h) - \Phi(x)}{\varepsilon}$$

$$= \tfrac{1}{2}\langle Ax, h\rangle + \tfrac{1}{2}\langle Ah, x\rangle - \langle b, h\rangle$$

$$= \langle Ax - b, h\rangle$$

$$= -\langle r, h\rangle$$

in other words

$$\nabla\Phi(x) = -r.$$

Gradient method with optimal stepping: Given an approximation x_N, define

$$x_{N+1} = \operatorname{argmin}\left\{ s\in\mathbb{R} \longmapsto \Phi(x_N - s\nabla\Phi(x_N)) = \Phi(x_N + s r_N) \right\}$$

$$\Phi(x_N + s r_N) = \frac{1}{2}\langle A x_N + s A r_N , x_N + s r_N\rangle - \langle b, x_N + s r_N\rangle$$

$$= \Phi(x_N) + s\langle \nabla\Phi(x_N), r_N\rangle + \frac{1}{2} s^2 \langle A r_N, r_N\rangle$$

$$= \Phi(x_N) - s\,\|r_N\|^2 + \frac{1}{2} s^2 \|r_N\|_A^2$$

minimized for $s = + \dfrac{\|r_N\|^2}{\|r_N\|_A^2}$.

<u>Hence</u>

$$\begin{cases} x_{N+1} = x_N + \dfrac{\|r_N\|^2}{\|r_N\|_A^2} r_N \\ r_{N+1} = b - A x_{N+1} \\ \quad = b - A x_N - \dfrac{\|r_N\|^2}{\|r_N\|_A^2} A r_N \\ \quad = r_N - \dfrac{\|r_N\|^2}{\|r_N\|_A^2} A r_N \end{cases}$$

<u>Note</u> Each iterate involves only one matrix vector product: $A r_N$ (both to compute $\|r_N\|_A^2$ and to update r_{N+1}).

<u>Convergence</u>

- $$x_{N+1} = x_N + \frac{\|r_N\|^2}{\|r_N\|_A^2} r_N$$

hence $e_{N+1} = e_N - \dfrac{\|r_N\|^2}{\|r_N\|_A^2} r_N$

- $$\langle r_{N+1}, r_N \rangle = \langle r_N, r_N \rangle - \frac{\|r_N\|^2}{\|r_N\|_A^2} \langle A r_N, r_N \rangle$$
$$= \|r_N\|^2 - \frac{\|r_N\|^2}{\|r_N\|_A^2} \|r_N\|_A^2 = 0 .$$

- $$\|e_{N+1}\|_A^2 = \langle A e_{N+1}, e_{N+1} \rangle$$
$$= \langle r_{N+1}, e_{N+1} \rangle$$
$$= \langle r_{N+1}, e_N - \frac{\|r_N\|^2}{\|r_N\|_A^2} r_N \rangle$$
$$= \langle r_{N+1}, e_N \rangle$$
$$= \langle r_N - \frac{\|r_N\|^2}{\|r_N\|_A^2} A r_N, e_N \rangle$$
$$= \|e_N\|_A^2 - \frac{\|r_N\|^4}{\|r_N\|_A^2}$$

$$\langle A r_N, e_N \rangle = \langle r_N, A e_N \rangle = \langle r_N, r_N \rangle = \|r_N\|^2$$

<u>Note</u> $\|e_N\|_A^2 = \langle A e_N, e_N \rangle = \langle r_N, A^{-1} r_N \rangle = \|r_N\|_{A^{-1}}^2$

$$\|e_{N+1}\|_A^2 = \left(1 - \frac{\|r_N\|^4}{\|r_N\|_A^2 \|r_N\|_{A^{-1}}^2}\right) \|e_N\|_A^2 .$$

If $\quad z = \sum_{i=1}^{N} \alpha_i e_i$

↑ orthonormal basis of eigenvectors of A with eigenvalues $0 < \lambda_1 \leq \cdots \leq \lambda_N$

$$\|z\|^4 = \left(\sum \alpha_i^2\right)^2$$

$$\|z\|_A^2 = \sum \lambda_i \alpha_i^2$$

$$\|z\|_{A^{-1}}^2 = \sum \frac{1}{\lambda_i} \alpha_i^2$$

$$\frac{\|z\|^4}{\|z\|_A^2 \|z\|_{A^{-1}}^2}$$ minimized by taking $\alpha_1^2 = \alpha_N^2 = 1/2$, $\alpha_j = 0 \quad 2 \leq j \leq N-1$

$$\rightsquigarrow \quad \|e_{N+1}\|_A^2 \leq \left(1 - \frac{4\lambda_{min}\lambda_{max}}{(\lambda_{min}+\lambda_{max})^2}\right) \|e_N\|_A^2$$

$$= \left(\frac{\lambda_{min} - \lambda_{max}}{\lambda_{min} + \lambda_{max}}\right)^2 \|e_N\|_A^2$$

$$= \left(\frac{K-1}{K+1}\right)^2 \|e_N\|_A^2$$

where $K = \frac{\lambda_{max}}{\lambda_{min}}$ is the CONDITION NUMBER of A.

$$\boxed{\begin{aligned} \|e_N\|_A &\leq \left(1 - \frac{2}{K+1}\right) \|e_{N-1}\|_A \\ &\leq \left(1 - \frac{2}{K+1}\right)^N \|e_0\|_A \end{aligned}}$$

<u>Recall</u> $\left(1 - \frac{\lambda}{k}\right)^k \simeq e^{-\lambda}$ as $k \to +\infty$, convergence requires $O(K)$ iterates.

Note that for the gradient method we have

$$x_n \in K_n = \text{span}\langle b, Ab, \dots, A^{n-1} b \rangle$$

↑ Krylov subspace

but each x_{n+1} is built using only x_n and the residual r_n at x_n.

For the conjugate gradient method, we instead rely on a global (on K_n) minimization:

$$x_0 = 0$$

$$x_n = \underset{x \in K_n}{\arg\min} \left\{ \Phi(x) = \frac{1}{2}\langle Ax, x\rangle - \langle b, x\rangle \right\}.$$

The key point is that x_n can be recursively constructed (along r_n and a sequence of line search directions p_n replacing the ~~p_n~~ direction r_n of the Gradient method).

Let us derive first some properties of the sequence (x_n).

- $\forall v_n \in K_n$

$$\frac{d}{d\varepsilon} \Phi(x_n + \varepsilon v_n)\Big|_{\varepsilon=0} = 0$$

$$\Rightarrow \langle r_n, v_n \rangle = 0$$

In other words

$$\underbrace{r_n}_{\in K_{n+1}} \perp K_n \quad \forall n \geq 1$$

and $r_0 = b \in K_1$.

$$\boxed{\text{Hence: as long as } r_n \neq 0, \text{ the sequence } (r_0, r_1, \dots, r_n) \text{ forms an orthogonal basis of } K_{n+1}.}$$

- Set $q_n := x_{n+1} - x_n$, i.e. $x_{n+1} = x_n + q_n$.

Therefore

$$r_{n+1} = r_n - Aq_n.$$

Note that $q_\ell \in K_{\ell+1} \quad \forall \ell \geq 0$

Therefore

$$\langle Aq_n, q_\ell \rangle = \langle r_n - r_{n+1}, \underbrace{q_\ell}_{\in K_n \text{ if } \ell < n} \rangle = 0 \text{ if } \ell < n$$

Hence : as long as $q_n \neq 0$, the sequence $(q_0, \dots, q_n)$ forms an A-orthogonal basis of K_{n+1}

The CG algorithm

$$x_0 = 0$$
$$e_0 = x_*$$
$$r_0 = b$$
$$p_0 = r_0 \ (= b).$$

1st: optimize Φ in $x_n + \mathbb{R} p_n$:

$$x_{n+1} = x_n + \alpha_{n+1} p_n$$

Which $\alpha_{n+1} \in \mathbb{R}$?

$$r_{n+1} = r_n - \alpha_{n+1} A p_n$$

$$0 = \langle r_{n+1}, r_n \rangle = \|r_n\|^2 - \alpha_{n+1} \langle A p_n, r_n \rangle$$

so must take $\alpha_{n+1} = \dfrac{\|r_n\|^2}{\langle A p_n, r_n \rangle}$

2nd : update search direction

$$p_{n+1} = r_{n+1} + \underbrace{\text{correction in } K_{n+1}}_{\text{it turns out we can take the correction in } \mathbb{R}\, p_n}$$

$$p_{n+1} = r_{n+1} + \beta_{n+1} p_n$$

We must (and it suffices) choose β_{n+1} so that

$$\langle A p_{n+1}, p_n \rangle = 0.$$

$$0 = \langle A p_{n+1}, p_n \rangle = \langle p_{n+1}, A p_n \rangle$$

$$= \left\langle r_{n+1} + \beta_{n+1} p_n, \frac{r_n - r_{n+1}}{\alpha_{n+1}} \right\rangle$$

$$= \frac{1}{\alpha_{n+1}} \left[- \|r_{n+1}\|^2 + \beta_{n+1} \langle p_n, r_n \rangle \right]$$

i.e. take $\beta_{n+1} = \dfrac{\|r_{n+1}\|^2}{\langle p_n, r_n \rangle}$

<u>Rmk</u> $p_n - r_n \in K_n$ hence

$$\alpha_{n+1} = \frac{\|r_n\|^2}{\|p_n\|_A^2}$$

$$\beta_{n+1} = \frac{\|r_{n+1}\|^2}{\|r_n\|^2}$$

<u>CG</u>

$$x_0 = 0$$
$$r_0 = b$$
$$p_0 = r_0$$

As long as $r_n \neq 0$ (or in practice a stopping threshold)

$$\alpha_{n+1} := \frac{\|r_n\|^2}{\|p_n\|_A^2} \longleftarrow \text{one MVP: } Ap_n$$

update x_n $\longrightarrow$ $x_{n+1} := x_n + \alpha_n p_n$

update r_n $\longrightarrow$ $r_{n+1} := r_n - \alpha_n A p_n$ re use

$$\beta_{n+1} := \frac{\|r_{n+1}\|^2}{\|r_n\|^2}$$

update p_n $\longrightarrow$ $p_{n+1} = r_{n+1} + \beta_{n+1} p_n$

<u>A second variational characterization</u>

Note that

$$\begin{aligned} e_n &= x_* - x_n \\ &= x_* - P_{n-1}(A)b \quad \text{where } P_{n-1} \in \mathbb{P}_{n-1}[X] \\ &= x_* - P_{n-1}(A) \cdot A x_* \\ &= Q_n(A) x_* = Q_n(A)\, e_0 \end{aligned}$$

where $Q_n \in \{ P \in \mathbb{P}_n[X] \text{ s.t. } P(0) = 1 \}$.

Hence,

$$\|e_n\|_A = \inf \{ \|Q_n(A) e_0\|_A , \ Q_n \in \mathbb{P}_n[X], Q(0)=1 \}$$

$$\leq \inf \left\{ \max_{\lambda \in \sigma(A)} |Q_n(\lambda)| , \ Q_n \in \mathbb{P}_n[X], Q_n(0)=1 \right\} \cdot \|e_0\|_A .$$

Here A is SPD, $\sigma(A) \subseteq [\lambda_{min}, \lambda_{max}]$.

Let $T_n(X)$ be the Tchebychev polynomial of degree n

<u>Recall</u> T_n characterized by

(*) $\quad T_n(\cos\theta) = \cos(n\theta) \quad \forall \theta \in \mathbb{R}$

It satisfies $|T_n(x)| \leq 1 \quad \forall x \in [-1, 1]$

and $T_n\left(\frac{z + z^{-1}}{2}\right) = \frac{z^n + z^{-n}}{2}$

[indeed, this is true for $z = e^{i\theta}$ by (*), and then everywhere on $\mathbb{C}\setminus\{0\}$ by eq. of polynomials, after mult. by z^n]

The affine map $g(x) = \dfrac{(\lambda_{max} + \lambda_{min}) - 2x}{\lambda_{max} - \lambda_{min}}$

is s.t. $g([\lambda_{min}, \lambda_{max}]) = [-1, 1]$

Set $Q_n(x) = \dfrac{P_n(g(x))}{P_n(g(0))} \in \mathbb{P}_n[x]$, $Q_n(0) = 1$

Then $\max\limits_{\lambda \in \sigma(A)} |Q_n(\lambda)| \leq \dfrac{1}{|P_n(g(0))|} \max\limits_{y \in [-1,1]} |P_n(y)|$

$$= \frac{1}{T_n\left(\frac{K+1}{K-1}\right)}$$

Note that

$\dfrac{K+1}{K-1} = \dfrac{z + z^{-1}}{2}$ for z s.t. $z^2 - 2\dfrac{K+1}{K-1} z + 1 = 0$

i.e. $z = \dfrac{K+1}{K-1} \pm \sqrt{\left(\dfrac{K+1}{K-1}\right)^2 - 1}$

$$= \frac{K+1}{K-1} \pm \sqrt{\frac{4K}{(K-1)^2}} = \frac{K+1 \pm 2\sqrt{K}}{(K-1)}$$

$$= \frac{(\sqrt{K} \pm 1)^2}{(\sqrt{K} - 1)(\sqrt{K} + 1)} = \frac{\sqrt{K} \pm 1}{\sqrt{K} \mp 1}$$

hence

$$T_n\left(\frac{K+1}{K-1}\right) = \frac{\left(\frac{\sqrt{K}+1}{\sqrt{K}-1}\right)^n + \left(\frac{\sqrt{K}+1}{\sqrt{K}-1}\right)^{-n}}{2}$$

$$\simeq \left(1 + \frac{2}{\sqrt{K}}\right)^n$$

<u>Finally</u> :

$$\| e_n \|_A \leq \frac{2}{\left(\frac{\sqrt{K}+1}{\sqrt{K}-1}\right)^n + \left(\frac{\sqrt{K}+1}{\sqrt{K}-1}\right)^{-n}} \| e_0 \|_A .$$

<u>Conclusion</u> Convergence requires $O(\sqrt{K})$ iterations.

<u>Additional remarks</u>

If $\#(\sigma(A)) = K \quad (\leq N)$

then CG converges in (at most) K iterations.

In particular CG has always converged after at most N iterations.

In practice $O(\sqrt{K}) \ll N$ and one never does that many iterations (finite precision arithmetics would anyway ruin this effort)